

PAULO T. FESALBON

AI ENGINEER

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PROFESSIONAL SUMMARY

A Computer Engineering graduate from the University of Batangas — Lipa Campus, Cum Laude and Top 1 Performing Student in my class. I am actively seeking opportunities as an AI Engineer, AI Automation Engineer, or in other AI-related roles where I can further develop my technical skills and knowledge. During my college years, I actively participated in various competitions involving both hardware and software projects. One of my proudest achievements was competing in the Philippine Innovation Conference (PIC) Hackathon, where my team placed 2nd out of more than 170 participating students, and leading my thesis team's AI-powered voice assistant, GISMO, to Best Thesis Overall across the entire College of Engineering and Architecture. Now I'm ready to bring that same drive into the industry as an AI Engineer or AI Automation Engineer, building real-world solutions and growing every step of the way.

AREAS OF EXPERTISE

• AI Systems • RAG, LangChain, NLP • Python Programming • Workflow Automation • ML Pipelines, Hugging Face

PROFESSIONAL EXPERIENCE

Solutions Intern Jun 2025 – Aug 2025

Procurement and Supply Institute of Asia (PASIA), Makati City, Philippines

- Trained and deployed a logistic regression model for automated spend categorization, leveraging historical procurement data (Description, Supplier Name, Categorization features) to streamline classification and reduce manual review time.
- Developed a Python script to automate email delivery via Gmail's SMTP server, supporting bulk recipient lists, secure environment-variable-based authentication, and built-in error handling — reducing manual send effort and eliminating failed-delivery blind spots.
- Onboarded onto data standardization workflows, completing 320 internship hours with a 90% performance rating.

Freelance 3D Modeler, Product Developer and Thesis Fabricator Jan 2024 – Present

Self Employed, Malvar, Batangas

- Designed and built AI-integrated systems combining machine learning models with hardware components (sensors, microcontrollers, embedded circuits), delivering functional prototypes for 9+ client and thesis projects across multiple schools.
- Developed and deployed AI-driven solutions — from concept to working prototype — integrating software intelligence with physical hardware through custom circuit design, fabrication, and 3D printing/modeling.
- Managed client-facing freelance and thesis projects for students beyond my own institution, applying AI integration expertise on select projects among 9+ overall and 3 thesis defenses, with consistent positive outcomes.

Technology Support Volunteer Jan 2020 – Present

Mt. Zion Fundamental Baptist Church, Santo Tomas, Batangas

- Operated OBS Studio to livestream weekly worship services on Facebook Live, ensuring consistent, high-quality broadcast delivery to the congregation and remote viewers.
- Managed stream setup, scene transitions, and audio/video quality in real time, minimizing technical disruptions during live services.
- Supported church technology needs on a volunteer basis for over 4 years, contributing to consistent digital outreach and service accessibility for the community.

EDUCATION

University of Batangas Lipa Campus

Jan 2022 – Jan 2026

BS Computer Engineering

Developed a strong technical foundation in AI, embedded systems, software development, automation, and hardware integration through hands-on engineering projects and applied coursework.

Recipient of the Outstanding Academic Achievement Award, Innovation and Technology Award, and Engineering Design Excellence Award.

TECHNICAL PROFICIENCIES

Programming Languages:	Python, Java, C, C++, MySQL, JavaScript, CSS, React
AI & Machine Learning:	LLM Applications, RAG, LangChain, NLP, Computer Vision (YOLOv8), Transformer Models, Hugging Face, STT, TTS, PyTorch
Automation & Integration:	Workflow Automation, CRM Systems (GoHighLevel, n8n), REST API Development, Third-Party API Integration
Networking & Security:	Cisco CCNAv7, Ethical Hacking, Digital Safety & Security

LINKS

- **My Portfolio:** <https://oluaps12.github.io/myportfolio>
- **Start-up:** likhaph.com
- **LinkedIn:** <https://www.linkedin.com/in/fesalbon-paulo-t-456193358/>

THESIS PROJECT

Lead AI Systems Developer

Jan 2024 – Aug 2026

Batangas

Design and Implementation of GISMO (General Information System for Multipurpose Outreach): An AI-Powered Information Assistant Using ESP32-S3

- Designed and implemented an AI-powered voice-activated assistant using ESP32-S3 as a client device connected to a local server, delivering 24/7 instant access to official university information and reducing administrative workload at UB Lipa Campus.
- Developed an NLP and retrieval-augmented generation (RAG) pipeline to interpret user voice queries and generate context-aware responses grounded solely in approved university documents, achieving 96.4% response accuracy in testing.
- Integrated embedded hardware with server-side AI inference via WebSocket-based audio streaming and a local server architecture, validated through testing with 22 student participants that showed 98.2% reliability and 95.4% user satisfaction.

ACHIEVEMENTS

Tugon Lipa - Philippine Innovation Conference 2025 Hackathon, Batangas

- Co-developed Tugon Lipa, an AI-powered smart community platform enabling Lipa City residents to report, resolve, and track local issues through real-time barangay social feeds and updates in a single accessible interface.
- Built an AI-driven reporting system with a user-friendly interface to close communication and responsiveness gaps between citizens and barangay-level governance.
- Competed at the "Hack the Future: Smart Batangas Province Hackathon Challenge" during the Philippine Innovation Conference 2025, securing 2nd Place among entries from students across the province, under the mentorship of Engr. Oscar Bryan Magtibay.

2026 BICPES Annual Mazebot Competition - Champion, Lipa City

- Engineered an autonomous maze-solving robot from scratch, designing a fully 3D-printed chassis and building the circuit hardware in-house around an ESP32 microcontroller and IR sensors for wall detection.
- Programmed a right-hand wall-following algorithm enabling the robot to autonomously navigate and solve the maze without human intervention.
- Won 1st Place (Champion) at the BICPES Annual Mazebot Competition 2026, completing the maze in 1 minute 35 seconds — the fastest time among all competing teams.

Best Paper (Overall) — CENAR Forum 2026: Awarded the highest research distinction for GISMO, an AI-powered virtual assistant, among all engineering research presentations.

Best Paper — 2nd Institutional Student Research Colloquium 2026: Recognized as the top research paper across all academic disciplines at the University of Batangas.

Best Paper — Computer Engineering Program to the Cenar Student Research Forum 2026: Received the highest research award within the Computer Engineering program for the GISMO AI Assistant project.

Best Presenter — 2nd Institutional Student Research Colloquium 2026: Recognized for delivering an outstanding research presentation of the GISMO AI Assistant at the University of Batangas Lipa Campus.

Outstanding Academic Achievement Award: For demonstrating exceptional scholastic performance, dedication to learning, and a consistent pursuit of intellectual excellence.

Innovation And Technology Award: For pioneering creative solutions and showcasing exceptional forward-thinking skills in applying technological advancements.

Engineering Design Excellence Award: For showcasing outstanding technical ingenuity, meticulous planning, and superior execution in engineering concepts.

Dean's Lister — University of Batangas: Maintained consistent academic excellence from the second year of the Computer Engineering program to the present.